CCE-Unit-3

I

Title :-

Grocery shopping list management using singly linked list C++

1) problem statement

Managing grocery items manually can be difficult when item need to be added. display searched or deleted frequently. A simple meny-driven application is required to maintain grocery item efficiently using a singly linked list.

The system should allow the user To add multiple grocery items. display The Complet shopping list. search for a particulor item. add delete an item when it is no longer required.

objectives.

- maintain grocery item using a singly L.L
- Add multiple grocery item dynamically.
- Display all items in The list.
- search for a particular grocery item.
- Delete an unwanted grocery item.
- understand practical linked list operations.

main operations:

Add items :- Create nodes and insert at the end.

Display list :- Traverse and display all item.

search item :- searched for a required item

Delete item :- Remove the selected item and adjust links.

Exit :- Terminate the program.

* Algorithm

1) start the program

2) Initialize head = NULL

3) Display the menu: Add items. Display list search item. Delete item. exit

4) Read the user choice

5) for choice 1: read the Number of item and insert each item at the end of the linked list.

6) For choice 2: check whether the list is empty. If not. Traverse and display all items.

7) For choice 3: read the item, Traverse The list. and compare each node. Display found or not found.

8) For choice 4: read the item. search for its node. adjust. delete The node. and display The result.

9) For choice 5: display The exit message and terminate.

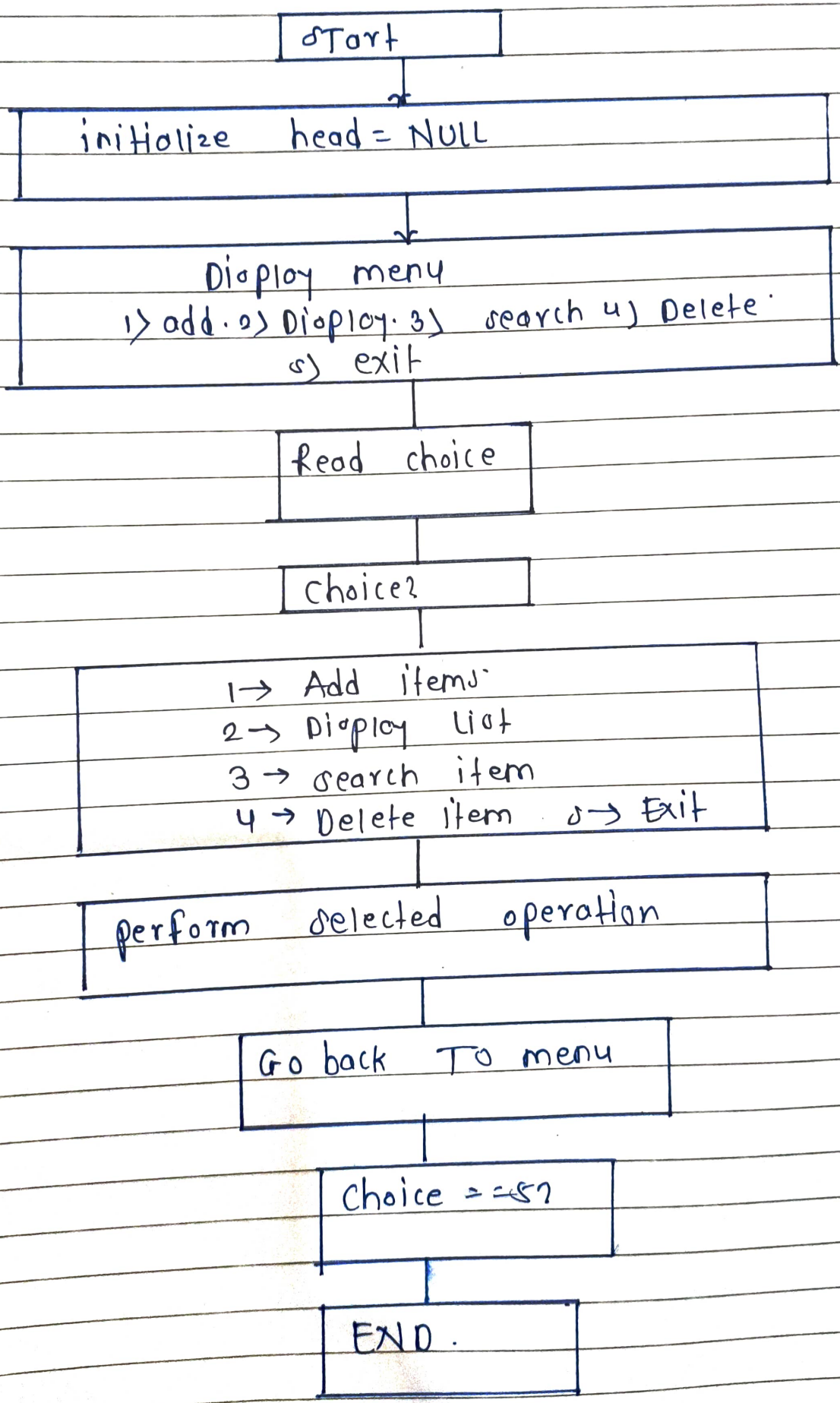
10) For any other choice. display an invalid choice message.

11) Repeat The menu until choice 5 is selected.

12) stop The program.

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flowchart



CCE- Unit 4

(4)

```
#include <iostream>
#include <string>
Using namespace std;

Struct Node
{
  String item;
  Node* next;
};

Node* head = NULL;

// Add multiple grocery items
Void addItem()
{
  Int n;

  Cout << "How many grocery items do you want to add? ";
  Cin >> n;

  For (int I = 0; I < n; i++)
  {
    Node* newNode = new Node;

    Cout << "Enter grocery item " << I + 1 << ": ";
    Cin >> newNode->item;
```

```
        newNode->next = NULL;

        if (head == NULL)
        {
           Head = newNode;
        }
        Else
        {
           Node* temp = head;

           While (temp->next != NULL)
           {
              Temp = temp->next;
           }

           Temp->next = newNode;
        }
     }

     Cout << "Items added successfully.\n";
}

// Display grocery list
Void displayList()
{
```

```
    If (head == NULL)
    {
      Cout << "Shopping list is empty.\n";
      Return;
    }

    Node* temp = head;

    Cout << "\nGrocery Shopping List:\n";

    While (temp != NULL)
    {
      Cout << "- " << temp->item << endl;
      Temp = temp->next;
    }
}

// Search grocery item
Void searchItem()
{
  String item;

  Cout << "Enter item to search: ";
  Cin >> item;

  Node* temp = head;
```

```
While (temp != NULL)
{
  If (temp->item == item)
  {
    Cout << "Item found in the shopping list.\n";
    Return;
  }

   Temp = temp->next;
 }

 Cout << "Item not found.\n";
}

// Delete grocery item
Void deleteItem()
{
  String item;

  Cout << "Enter item to delete: ";
  Cin >> item;

  Node* temp = head;
  Node* prev = NULL;
```

```
While (temp != NULL && temp->item != item)
{
   Prev = temp;
   Temp = temp->next;
 }

 If (temp == NULL)
 {
    Cout << "Item not found.\n";
    Return;
 }

 If (prev == NULL)
 {
    Head = temp->next;
 }
 Else
 {
    Prev->next = temp->next;
 }

  Delete temp;

  Cout << "Item deleted successfully.\n";
}
```

```
Int main()
{
  Int choice;

  Do
  {
    Cout << "\n===== Grocery Shopping List =====\n";
    Cout << "1. Add Items\n";
    Cout << "2. Display List\n";
    Cout << "3. Search Item\n";
    Cout << "4. Delete Item\n";
    Cout << "5. Exit\n";

    Cout << "Enter your choice: ";
    Cin >> choice;

    Switch (choice)
    {
      Case 1:
        addItem();
        break;

      case 2:
        displayList();
        break;
```

```
        case 3:
            searchItem();
            break;

        case 4:
            deleteItem();
            break;

        case 5:
            cout << "Exiting program...\n";
            break;

        default:
            cout << "Invalid choice! Please try again.\n";
    }

} while (choice != 5);

Return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS D:\siddanth> cd 'd:\siddanth\output'
PS D:\siddanth\output> & .\'liked list 3.exe'

===== Grocery Shopping List =====
1. Add Items
2. Display List
3. Search Item
4. Delete Item
5. Exit
Enter your choice: 1
How many grocery items do you want to add? 4
Enter grocery item 1: sugar
Enter grocery item 2: corn
Enter grocery item 3: rice
Enter grocery item 4: wheat
Items added successfully.

===== Grocery Shopping List =====
1. Add Items
2. Display List
3. Search Item
4. Delete Item
5. Exit
Enter your choice: 2

Grocery Shopping List:
- sugar
- corn
- rice
- wheat

===== Grocery Shopping List =====
1. Add Items
2. Display List
3. Search Item
4. Delete Item
5. Exit
Enter your choice: 3
Enter item to search: sugar
```

Problems Output BlameConsole Terminal Ports

```
Grocery Shopping List
1. Add Items
2. Display List
3. Search Item
4. Delete Item
5. Exit
Enter your choice: 3
Enter item to search: eggs
Item found in the shopping list.

Grocery Shopping List
1. Add Items
2. Display List
3. Search Item
4. Delete Item
5. Exit
Enter your choice: 4
Enter item to delete: bread
Item deleted successfully.

Grocery Shopping List
1. Add Items
2. Display List
3. Search Item
4. Delete Item
5. Exit
Enter your choice: 2

Grocery Shopping List
1. eggs
2. milk
3. rice

Grocery Shopping List
1. Add Items
2. Display List
3. Search Item
4. Delete Item
5. Exit
```